

# Exam Students Score mini report

## Part - A theory

1) What is mean, median and mode with a real life example?

ans mean is the average of all values.  
median is the middle value when data is arranged in order.  
mode is the value that occurs more frequently

Example :  $\{1, 2, 3, 3, 4, 8\}$

mean = 3.5

median = 3

mode = 3

2) What is the difference between variance and Standard deviation?

ans variance measure how far data values are spread from the mean using squared units.

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Standard deviation is the square root of variance and is expressed in the same units as the original data.

3 Define Normal distribution and give one practical use case?

ans A normal distribution is a bell shaped curve distribution most values are near the mean. ~~and~~ ~~it is~~  
It is symmetric around the mean.

Example: Schools can use this distribution to identify average performance and compare a student score with the overall class performance.

4) Explain Skewness and Kurtosis ?

ans Skewness : Skewness tell us whether data is Symmetric or has a longer tail on one side.

Kurtosis : Kurtosis describe the heaviness of the tails or the Presence of extreme values.

5) What is Probability ?

ans Probability measures how likely an event is to happen, with a value between 0 and 1.

★ Difference between

theoretical Probability

⇒ Theoretical Probability

is based on expected outcomes.

Empirical Probability

Empirical Probability

is based on actual observations or experiments.

6) Explain Independent & dependent events?

ans Independent events are events where one event does not affect the Probability of another.

Example : tossing a coin twice.

A ~ Head

B ~ Tail

So, Head A does not affect Tail B.

Dependent events are events where the first event affects the Probability of the Second

Example : Drawing 2 card without replacement.

7 What is the intuition of Bayes' theorem?

ans Bayes theorem helps to update Probability of event when new evidence is available.

Example : If a medical test is positive Bayes' theorem helps determine the Probability of the person actually has disease based on test accuracy and disease Prevalance.

8) Explain Eigenvalue and eigenvector?  
ans Eigenvalue is the amount that tells how much that eigenvector is stretched or shrunk.

Eigenvector is a special vector whose direction does not change when a matrix transforms it.

Basic equation is  $Av = \lambda v$ .

Example :  $\lambda = 3$ , the eigenvector is stretched to three times its original length

The E.N.D